

Lecture Notes on Diffusion Processes

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Contents

1	Itô's Processes	2
1.1	Brownian Motion and Itô's Processes	2
1.2	Quadratic Variation	3
1.3	Itô's Chain Rule	6
1.4	Markov Semigroup	8
1.5	Forward Equation and Fokker–Planck Equation	10
2	Langevin Dynamics	13
2.1	Generator and stationary distribution	13
2.2	Dirichlet Form	14
3	Feynman–Kac Formula	17
3.1	Terminal-value problems	17
3.2	Boundary value problems and stopping times	22

1 Itô's Processes

This section develops the basic theory of Itô's processes. After introducing Brownian motion and Itô's integral, we establish the properties of Itô's integral needed to derive Itô's chain rule, and introduce the infinitesimal generator and its associated Markov semigroup. We conclude with the dual, forward equation for the law of the process. For background on stochastic calculus and Langevin diffusions, see [Särkkä and Solin, 2019] and [Chewi, 2024].

1.1 Brownian Motion and Itô's Processes

We begin by defining Brownian motion, Itô's integral, and Itô's process, and give a few basic examples of diffusions that will be revisited throughout these notes.

Let $(B_t)_{t \geq 0}$ be a Brownian motion over $\mathbb{R}$. B_t satisfies the following properties:

1. $B_0 = 0$ almost surely.
2. Independent increments: for any $0 \leq t_0 < t_1 < \dots < t_n$, the increments $B_{t_1} - B_{t_0}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent.
3. Stationary Gaussian increments: for $0 \leq s < t$, $B_t - B_s \sim \mathcal{N}(0, t - s)$.
4. Continuous paths: $t \mapsto B_t$ is almost surely continuous.

We denote by B_t a one-dimensional Brownian motion and by $\mathbf{B}_t$ a d -dimensional Brownian motion.

Given a function $g : \mathbb{R} \rightarrow \mathbb{R}$ and a process x_t , we define its Itô's integral as

$$\int_0^T g(x_t) dB_t := \lim_{|\pi| \rightarrow 0} \sum_{k=0}^{n-1} g(x_{t_k}) (B_{t_{k+1}} - B_{t_k}),$$

where $\pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$ is a partition of $[0, T]$, with mesh

$$|\pi| := \max_{0 \leq k \leq n-1} (t_{k+1} - t_k) \rightarrow 0.$$

We define an Itô's process as

$$x_T - x_0 = \int_0^T f(x_t) dt + \int_0^T g(x_t) dB_t.$$

We can also write the above integral expression in its derivative form as

$$dx_t = f(x_t) dt + g(x_t) dB_t.$$

We call f the drift and g the diffusion terms of the Itô's process.

Example 1 (Pure Brownian). Take $f \equiv 0$ and $g \equiv 1$ in the general Itô's process, so that

$$dx_t = dB_t.$$

In this case $x_t = B_t$ for every $t \in [0, T]$, and since $B_t \sim \mathcal{N}(0, t)$,

$$\mathbb{E}[x_t] = 0, \quad \text{Var}[x_t] = t.$$

For $0 \leq s \leq t$, writing $B_t = B_s + (B_t - B_s)$ and using that $B_t - B_s$ is independent of B_s with mean zero,

$$\text{Cov}(x_s, x_t) = \mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s(B_t - B_s)] = s = \min(s, t).$$

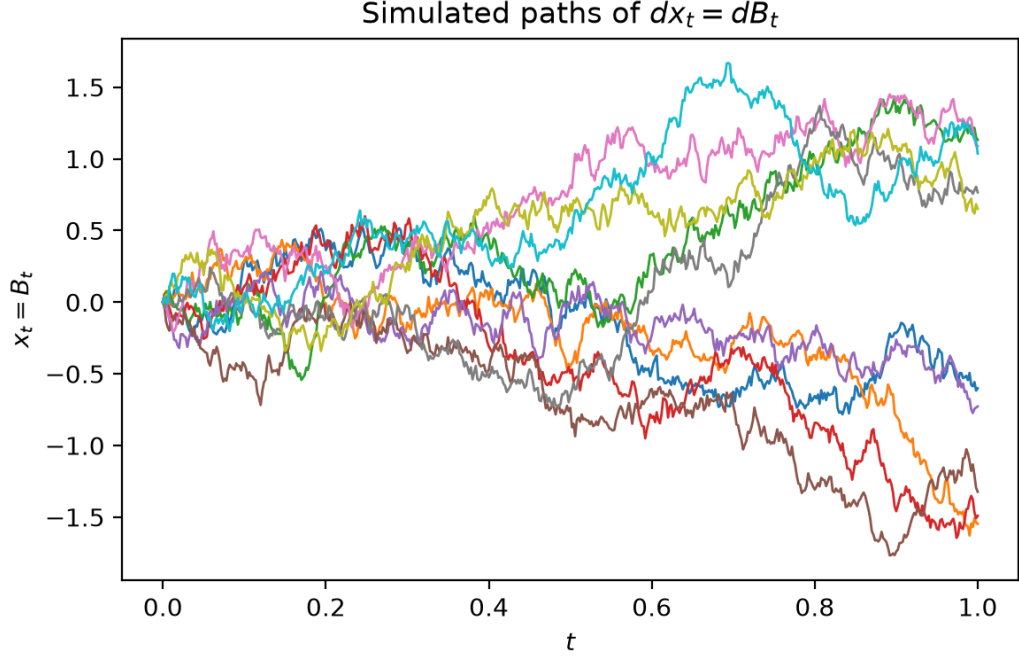


Figure 1: Simulated sample paths of a pure Brownian motion, $dx_t = dB_t$.

Example 2 (Ornstein-Uhlenbeck). Take $f(x) = -\theta x$ and $g \equiv \sigma$, for $\theta, \sigma > 0$, so that

$$dx_t = -\theta x_t dt + \sigma dB_t.$$

In integral form,

$$x_T = x_0 - \theta \int_0^T x_t dt + \sigma B_T.$$

Taking expectation and using that $\mathbb{E}[B_T] = 0$,

$$\mathbb{E}[x_T \mid x_0] = x_0 - \theta \int_0^T \mathbb{E}[x_t \mid x_0] dt.$$

Writing $m(T) := \mathbb{E}[x_T \mid x_0]$, this says m solves the ordinary differential equation $m'(T) = -\theta m(T)$ with $m(0) = x_0$, so

$$\mathbb{E}[x_T \mid x_0] = x_0 e^{-\theta T}.$$

We compute the variance and covariance of x_t in Example 7, once Itô's chain rule is available.

Example 3 (Langevin). Let π be a target density on $\mathbb{R}$ and let $\nabla \log \pi$ be its score. Take $f = \nabla \log \pi$ and $g \equiv \sqrt{2}$, so that

$$dx_t = \nabla \log \pi(x_t) dt + \sqrt{2} dB_t.$$

Unlike the previous two examples, f is nonlinear in general, so x_t has no closed form and we cannot compute $\mathbb{E}[x_t]$ or $\text{Var}[x_t]$ with the tools developed so far. We will study the long-run behavior of this process later in these notes, once we have the generator and the Fokker-Planck equation.

1.2 Quadratic Variation

In this section we establish two basic properties of Itô's integral driven by Brownian motion that will be needed to derive Itô's chain rule: its behavior under a telescoping sum, and its quadratic variation, which explains the informal identity $(dB_t)^2 = dt$ used throughout these notes.

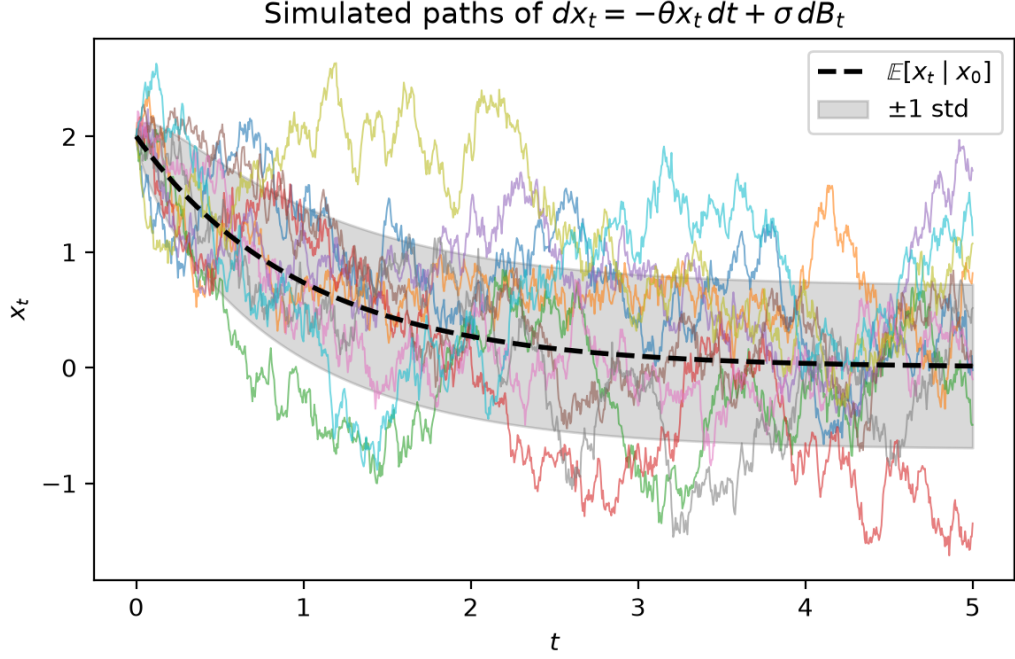


Figure 2: Simulated sample paths of an Ornstein-Uhlenbeck process, $dx_t = -\theta x_t dt + \sigma dB_t$, together with the mean $\mathbb{E}[x_t | x_0]$ and a ± 1 standard deviation band.

As a first example,

$$\int_0^T dB_t = B_T - B_0,$$

since

$$\sum_{k=0}^{n-1} (B_{t_{k+1}} - B_{t_k}) = B_T - B_0$$

by a telescopic sum.

A second important object is the sum of squared increments

$$\sum_{k=0}^{n-1} (B_{t_{k+1}} - B_{t_k})^2.$$

Unlike what happens for a smooth function, this quantity does not vanish as the mesh goes to zero. For Brownian motion it converges to T . This motivates the definition of the *quadratic variation*

$$[B]_T := \lim_{|\pi| \rightarrow 0} \sum_{k=0}^{n-1} (B_{t_{k+1}} - B_{t_k})^2.$$

To see why this limit is T , define $\Delta B_k := B_{t_{k+1}} - B_{t_k}$ and $\Delta t_k := t_{k+1} - t_k$. Since $\Delta B_k \sim \mathcal{N}(0, \Delta t_k)$, we have $\mathbb{E}[(\Delta B_k)^2] = \Delta t_k$. Therefore,

$$\mathbb{E} \left[\sum_{k=0}^{n-1} (\Delta B_k)^2 \right] = \sum_{k=0}^{n-1} \Delta t_k = T.$$

Moreover, if $X \sim \mathcal{N}(0, \sigma^2)$, a standard computation (e.g. differentiating the moment generating function $\mathbb{E}[e^{\lambda X}] = e^{\lambda^2 \sigma^2 / 2}$ four times at $\lambda = 0$) gives the fourth moment $\mathbb{E}[X^4] = 3\sigma^4$. Applying

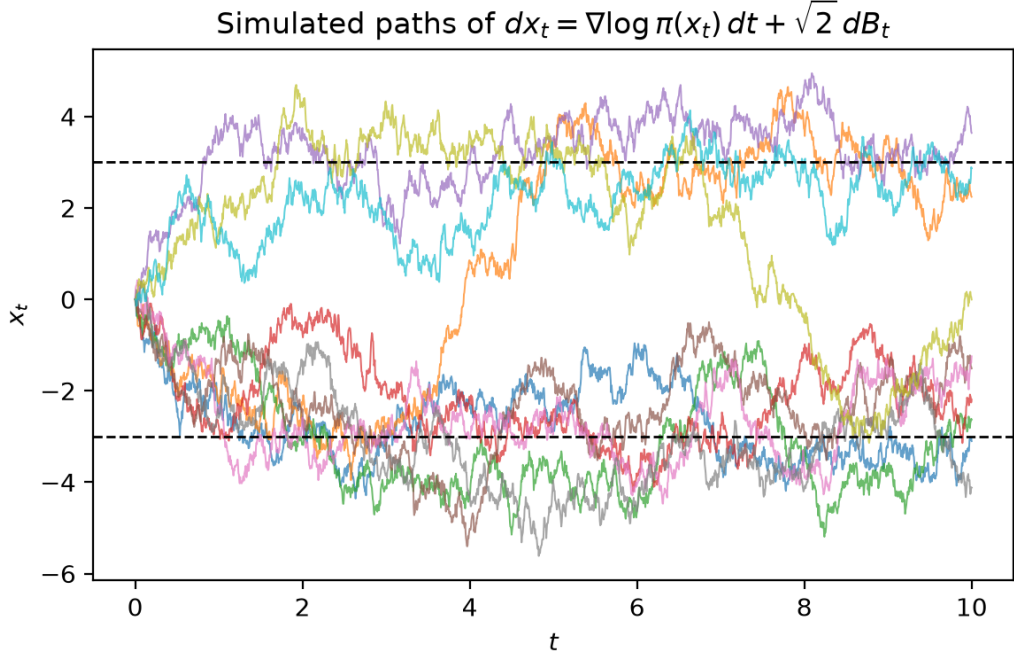


Figure 3: Simulated paths of $dx_t = \nabla \log \pi(x_t) dt + \sqrt{2} dB_t$ for π a mixture of two Gaussians with means $\mu = \pm 3$ (dashed lines).

the identity $\text{Var}[Y] = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$ to $Y = X^2$, so that $\mathbb{E}[Y] = \mathbb{E}[X^2] = \sigma^2$ and $\mathbb{E}[Y^2] = \mathbb{E}[X^4]$, gives

$$\text{Var}[X^2] = \mathbb{E}[X^4] - (\mathbb{E}[X^2])^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4.$$

Since $\Delta B_k \sim \mathcal{N}(0, \Delta t_k)$, taking $\sigma^2 = \Delta t_k$ gives $\text{Var}[(\Delta B_k)^2] = 2(\Delta t_k)^2$.

Because Brownian increments over disjoint intervals are independent, the random variables $(\Delta B_0)^2, \dots, (\Delta B_{n-1})^2$ are independent, so all their pairwise covariances vanish and the variance of their sum is the sum of their variances:

$$\text{Var}\left[\sum_{k=0}^{n-1} (\Delta B_k)^2\right] = \sum_{k=0}^{n-1} \text{Var}[(\Delta B_k)^2] = 2 \sum_{k=0}^{n-1} (\Delta t_k)^2.$$

Bounding each factor Δt_k by the mesh $|\pi| = \max_k \Delta t_k$, and using that the partition telescopes, $\sum_{k=0}^{n-1} \Delta t_k = T$,

$$2 \sum_{k=0}^{n-1} (\Delta t_k)^2 \leq 2|\pi| \sum_{k=0}^{n-1} \Delta t_k = 2T|\pi| \rightarrow 0$$

as $|\pi| \rightarrow 0$. Therefore,

$$\sum_{k=0}^{n-1} (\Delta B_k)^2 \rightarrow T$$

in L^2 , and hence also in probability. Thus $[B]_T = T$.

It is common to write this result formally as

$$\int_0^T (dB_t)^2 = \int_0^T dt = T,$$

or, in differential notation, as $(dB_t)^2 = dt$.

1.3 Itô's Chain Rule

In this section we derive Itô's chain rule, which describes how to differentiate a function of a diffusion process. Using the definitions in the previous section, we can show that the usual chain rule does not work in Itô's calculus. Indeed,

$$B_{t_{k+1}}^2 - B_{t_k}^2 = 2B_{t_k} \Delta B_k + (\Delta B_k)^2.$$

Summing over k ,

$$B_T^2 - B_0^2 = 2 \sum_{k=0}^{n-1} B_{t_k} \Delta B_k + \sum_{k=0}^{n-1} (\Delta B_k)^2.$$

As $|\pi| \rightarrow 0$, the first term converges to the Itô integral $\int_0^T B_t dB_t$, while the second converges to $[B]_T = T$. Therefore,

$$B_T^2 - B_0^2 = 2 \int_0^T B_t dB_t + T.$$

In particular, if $B_0 = 0$,

$$\int_0^T B_t dB_t = \frac{1}{2}(B_T^2 - T).$$

In differential notation, this identity becomes

$$dB_t^2 = 2B_t dB_t + (dB_t)^2 = 2B_t dB_t + dt.$$

Thus, the additional dt term arises precisely because Brownian motion has nonzero quadratic variation.

We now derive how the chain rule behaves for a general Itô's process.

Consider a diffusion process in $\mathbb{R}$ satisfying

$$dx_t = f(x_t) dt + g(x_t) dB_t.$$

We want a formula for $d\phi(x_t)$ for a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Intuitively, note that

$$\phi(x_t + dx_t) = \phi(x_t) + \nabla \phi(x_t) dx_t + dx_t \frac{1}{2} \nabla^2 \phi(x_t) dx_t + \dots$$

Replacing dx_t by its expression:

$$\begin{aligned} d\phi(x_t) &= \nabla \phi(x_t) (f(x_t) dt + g(x_t) dB_t) \\ &\quad + (f(x_t) dt + g(x_t) dB_t) \frac{1}{2} \nabla^2 \phi(x_t) (f(x_t) dt + g(x_t) dB_t) + \dots \end{aligned}$$

Moreover, we expect that $dt dt = dt dB_t = 0$ and by the quadratic variation of Brownian motion, $dB_t dB_t = dt$, hence:

$$d\phi(x_t) = \left(\nabla \phi(x_t) f(x_t) + g(x_t) g(x_t) \frac{1}{2} \nabla^2 \phi(x_t) \right) dt + \nabla \phi(x_t) g(x_t) dB_t$$

A fully rigorous proof would write Itô's process in integral form and repeat this argument over the partition used to define Itô's integral; the informal argument above is enough to grasp the idea. For a complete proof, see [Øksendal, 2003].

More generally, if the process is defined over $\mathbb{R}^d$, Itô's chain rule becomes:

Proposition 1 (Itô formula for the observable). *Let $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ be sufficiently regular. Then*

$$d\phi(\mathbf{x}_t) = \left[\nabla_{\mathbf{x}} \phi(\mathbf{x}_t)^\top f(\mathbf{x}_t) + \frac{1}{2} \text{tr} \left\{ \left(\nabla_{\mathbf{x}} \nabla_{\mathbf{x}}^\top \phi(\mathbf{x}_t) \right) L(\mathbf{x}_t) L(\mathbf{x}_t)^\top \right\} \right] dt + \nabla_{\mathbf{x}} \phi(\mathbf{x}_t)^\top L(\mathbf{x}_t) dB_t.$$

Example 4 (Brownian squared). Let $dx_t = dB_t$, i.e. $f \equiv 0$ and $g \equiv 1$, and take $\phi(x) = x^2$, so that $\nabla\phi(x) = 2x$ and $\nabla^2\phi(x) = 2$. By Itô's chain rule,

$$d(x_t)^2 = d(B_t)^2 = \left(2x_t \cdot 0 + \frac{1}{2} \cdot 1^2 \cdot 2\right)dt + 2x_t \cdot 1 dB_t = dt + 2B_t dB_t,$$

that is,

$$x_T^2 = T + \int_0^T B_t dB_t,$$

matching the identity $\int_0^T B_t dB_t = \frac{1}{2}(B_T^2 - T)$ derived earlier in this section directly from the definition of Itô's integral.

Example 5 (Geometric Brownian motion). Let $dx_t = dB_t$ (i.e. $f \equiv 0, g \equiv 1$) and take $\phi(x) = e^x$, so that $\nabla\phi(x) = \nabla^2\phi(x) = \phi(x)$. By Itô's chain rule,

$$d(e^{x_t}) = \left(0 + \frac{1}{2} \cdot 1^2 \cdot e^{x_t}\right)dt + e^{x_t} \cdot 1 dB_t = \frac{1}{2}e^{x_t} dt + e^{x_t} dB_t.$$

Writing $Y_t := e^{B_t}$, this says $dY_t = Y_t(\frac{1}{2}dt + dB_t)$, i.e. Y_t is a geometric Brownian motion. Unlike B_t itself, Y_t is not a martingale: it has a strictly positive drift $\frac{1}{2}Y_t dt$. The next example shows how to correct for this.

Example 6 (Exponential martingale). For a function $\phi(t, x)$ that also depends explicitly on t , the same heuristic as before gives an extra $\partial_t\phi dt$ term, since t itself has zero quadratic variation:

$$d\phi(t, x_t) = \left(\partial_t\phi(t, x_t) + f(x_t)\nabla\phi(t, x_t) + \frac{1}{2}g(x_t)^2\nabla^2\phi(t, x_t)\right)dt + g(x_t)\nabla\phi(t, x_t)dB_t,$$

where ∇, ∇^2 denote derivatives in x .

Take $\phi(t, x) = e^{x-t/2}$ and $M_t := \phi(t, B_t) = e^{B_t-t/2}$. Then $\partial_t\phi = -\frac{1}{2}\phi$ and $\nabla\phi = \nabla^2\phi = \phi$, so

$$dM_t = \left(-\frac{1}{2}M_t + 0 + \frac{1}{2} \cdot 1^2 \cdot M_t\right)dt + M_t \cdot 1 dB_t = M_t dB_t.$$

The $-t/2$ correction cancels exactly the drift found in the previous example: $M_t = e^{B_t-t/2}$ has no dt term in its Itô decomposition, i.e. zero drift, which is precisely the martingale property used below to define the generator.

Example 7 (Ornstein-Uhlenbeck revisited). We revisit Example 2 and compute the closed-form solution, variance, and covariance of $dx_t = -\theta x_t dt + \sigma dB_t$.

To remove the drift, take $\phi(t, x) = e^{\theta t}x$ in the time-dependent formula from Example 6, so $\partial_t\phi = \theta\phi$, $\nabla\phi = e^{\theta t}$, $\nabla^2\phi = 0$. Then

$$d(e^{\theta t}x_t) = \left(\theta e^{\theta t}x_t + e^{\theta t} \cdot (-\theta x_t) + 0\right)dt + e^{\theta t}\sigma dB_t = \sigma e^{\theta t} dB_t.$$

Integrating from 0 to t gives the closed form

$$x_t = x_0 e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dB_s,$$

recovering the mean $\mathbb{E}[x_t | x_0] = x_0 e^{-\theta t}$ found in Example 2.

Writing the integral as a limit of sums as in the definition of Itô's integral, $\sigma \int_0^t e^{-\theta(t-s)} dB_s = \lim_{|\pi| \rightarrow 0} \sigma \sum_k e^{-\theta(t-t_k)} \Delta B_k$, whose terms are independent with variance $\sigma^2 e^{-2\theta(t-t_k)} \Delta t_k$, so

$$\text{Var}[x_t | x_0] = \sigma^2 \int_0^t e^{-2\theta(t-s)} ds = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t}).$$

For $0 \leq s \leq t$, split the closed-form solution at s and use $t - r = (t - s) + (s - r)$ to factor:

$$\begin{aligned} x_t &= x_0 e^{-\theta t} + \sigma \int_0^s e^{-\theta(t-r)} dB_r + \sigma \int_s^t e^{-\theta(t-r)} dB_r \\ &= e^{-\theta(t-s)} \underbrace{\left(x_0 e^{-\theta s} + \sigma \int_0^s e^{-\theta(s-r)} dB_r \right)}_{=x_s} + \sigma \int_s^t e^{-\theta(t-u)} dB_u, \end{aligned}$$

where the underbraced term is exactly x_s , i.e. the closed form evaluated at s . The last term involves only Brownian increments after s , so it is independent of x_s (by the independent-increments property of Brownian motion) and has mean zero. Writing $Z := \sigma \int_s^t e^{-\theta(t-u)} dB_u$, so that $x_t = e^{-\theta(t-s)} x_s + Z$ with Z independent of x_s , bilinearity of the covariance gives

$$\text{Cov}(x_s, x_t \mid x_0) = e^{-\theta(t-s)} \text{Var}[x_s \mid x_0] + \underbrace{\text{Cov}(x_s, Z \mid x_0)}_{=0} = \frac{\sigma^2}{2\theta} e^{-\theta(t-s)} (1 - e^{-2\theta s}).$$

1.4 Markov Semigroup

The generator is the operator that appears after the martingale part of Itô's formula vanishes under conditional expectation.

Definition 1. The infinitesimal generator $\mathcal{A}$ is defined by

$$\mathcal{A}\phi(\mathbf{x}) := \lim_{\delta \rightarrow 0} \frac{\mathbb{E}[\phi(\mathbf{x}_{s+\delta}) \mid \mathbf{x}_s = \mathbf{x}] - \phi(\mathbf{x})}{\delta},$$

whenever this limit exists. By time homogeneity, the definition does not depend on s .

Proposition 2 (Expression for the generator). For the diffusion above,

$$\mathcal{A}\phi(\mathbf{x}_s) = \nabla_x \phi(\mathbf{x}_s)^\top f(\mathbf{x}_s) + \frac{1}{2} \text{tr} \left\{ \left(\nabla_x \nabla_x^\top \phi(\mathbf{x}_s) \right) L(\mathbf{x}_s) L(\mathbf{x}_s)^\top \right\}.$$

Proof. Integrating Itô's formula from s to t yields

$$\begin{aligned} \phi(\mathbf{x}_t) - \phi(\mathbf{x}_s) &= \int_s^t \left[\nabla_x \phi(\mathbf{x}_r)^\top f(\mathbf{x}_r) + \frac{1}{2} \text{tr} \left\{ \left(\nabla_x \nabla_x^\top \phi(\mathbf{x}_r) \right) L(\mathbf{x}_r) L(\mathbf{x}_r)^\top \right\} \right] dr \\ &\quad + \int_s^t \nabla_x \phi(\mathbf{x}_r)^\top L(\mathbf{x}_r) d\mathbf{B}_r. \end{aligned}$$

The stochastic integral

$$M_t := \int_0^t \nabla_x \phi(\mathbf{x}_r)^\top L(\mathbf{x}_r) d\mathbf{B}_r$$

is a martingale under the usual square-integrability assumptions. Therefore

$$\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] = 0.$$

Since $\mathbf{x}_s$ is $\mathcal{F}_s$ -measurable, the tower property implies

$$\mathbb{E}[M_t - M_s \mid \mathbf{x}_s] = \mathbb{E}[\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] \mid \mathbf{x}_s] = 0.$$

Thus, for $t = s + \delta$,

$$\begin{aligned} &\mathbb{E}[\phi(\mathbf{x}_{s+\delta}) \mid \mathbf{x}_s] - \phi(\mathbf{x}_s) \\ &= \left[\nabla_x \phi(\mathbf{x}_s)^\top f(\mathbf{x}_s) + \frac{1}{2} \text{tr} \left\{ \left(\nabla_x \nabla_x^\top \phi(\mathbf{x}_s) \right) L(\mathbf{x}_s) L(\mathbf{x}_s)^\top \right\} \right] \delta + o(\delta). \end{aligned}$$

Dividing by δ and taking $\delta \rightarrow 0$ gives the result. $\square$

The generator is naturally connected with the Markov semigroup. In this notation, we write

$$(P_t\phi)(\mathbf{x}) := \mathbb{E}[\phi(\mathbf{x}_t) \mid \mathbf{x}_0 = \mathbf{x}].$$

Then the generator can also be written as

$$\mathcal{A}\phi = \lim_{\delta \rightarrow 0} \frac{P_\delta\phi - P_0\phi}{\delta}.$$

Proposition 3 (Kolmogorov backward equation). *Let ϕ belong to the domain of $\mathcal{A}$, and assume that the semigroup preserves this domain. Then*

$$P_t\mathcal{A}\phi = \mathcal{A}P_t\phi = \frac{d}{dt}P_t\phi.$$

Proof. The definition of the generator and the semigroup property give

$$P_t\mathcal{A}\phi = \lim_{\delta \rightarrow 0} P_t \frac{P_\delta\phi - P_0\phi}{\delta} = \lim_{\delta \rightarrow 0} \frac{P_t P_\delta\phi - P_t P_0\phi}{\delta} = \lim_{\delta \rightarrow 0} \frac{P_{t+\delta}\phi - P_t\phi}{\delta} = \frac{d}{dt}P_t\phi.$$

Similarly,

$$\mathcal{A}P_t\phi = \lim_{\delta \rightarrow 0} \frac{P_\delta P_t\phi - P_0 P_t\phi}{\delta} = \frac{d}{dt}P_t\phi.$$

□

Under the usual semigroup assumptions, this evolution is written

$$P_t\phi = e^{t\mathcal{A}}P_0\phi = e^{t\mathcal{A}}\phi.$$

For a one-dimensional process, Proposition 2 reduces to

$$\mathcal{A}\phi(x) = f(x)\nabla\phi(x) + \frac{1}{2}g(x)^2\nabla^2\phi(x).$$

We revisit our three running examples.

Example 8 (Generator of Brownian motion). *For pure Brownian motion (Example 1), $f \equiv 0$ and $g \equiv 1$, so*

$$\mathcal{A}\phi(x) = \frac{1}{2}\nabla^2\phi(x),$$

i.e. the generator is half the Laplacian.

Example 9 (Generator of the Ornstein-Uhlenbeck process). *For the Ornstein-Uhlenbeck process (Example 2), $f(x) = -\theta x$ and $g \equiv \sigma$, so*

$$\mathcal{A}\phi(x) = -\theta x \nabla\phi(x) + \frac{\sigma^2}{2}\nabla^2\phi(x).$$

Example 10 (Generator of Langevin dynamics). *For the Langevin process (Example 3), $f = \nabla \log \pi$ and $g \equiv \sqrt{2}$, so*

$$\mathcal{A}\phi(x) = \nabla \log \pi(x) \nabla\phi(x) + \nabla^2\phi(x).$$

1.5 Forward Equation and Fokker–Planck Equation

This subsection derives the forward (Fokker–Planck–Kolmogorov) equation for the density of the process, dual to the backward equation of Proposition 3.

Assume that the law Q_t of $\mathbf{x}_t$ admits a density q_t with respect to Lebesgue measure, so that

$$Q_t(d\mathbf{x}) = q_t(\mathbf{x}) d\mathbf{x}.$$

The duality relation between the Markov semigroup and the law of the process then becomes

$$\int (P_t\phi)(\mathbf{x}) q_0(\mathbf{x}) d\mathbf{x} = \int \phi(\mathbf{x}) q_t(\mathbf{x}) d\mathbf{x}.$$

Thus, at the level of densities, the duality pairing is simply the usual integral pairing between functions.

Proposition 4 (Forward equation). *Suppose that Q_t admits a sufficiently regular density q_t . Then q_t satisfies*

$$\frac{\partial}{\partial t} q_t = \mathcal{A}^* q_t,$$

where $\mathcal{A}^*$ is the adjoint of $\mathcal{A}$, characterized by

$$\int (\mathcal{A}\phi)(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} = \int \phi(\mathbf{x}) (\mathcal{A}^* q)(\mathbf{x}) d\mathbf{x}$$

for sufficiently regular ϕ and q .

Proof. Starting from the duality relation

$$\int (P_t\phi)(\mathbf{x}) q_0(\mathbf{x}) d\mathbf{x} = \int \phi(\mathbf{x}) q_t(\mathbf{x}) d\mathbf{x},$$

we differentiate with respect to t . Using the backward equation

$$\frac{\partial}{\partial t} P_t\phi = P_t\mathcal{A}\phi,$$

we obtain

$$\begin{aligned} \frac{d}{dt} \int \phi(\mathbf{x}) q_t(\mathbf{x}) d\mathbf{x} &= \frac{d}{dt} \int (P_t\phi)(\mathbf{x}) q_0(\mathbf{x}) d\mathbf{x} \\ &= \int (P_t\mathcal{A}\phi)(\mathbf{x}) q_0(\mathbf{x}) d\mathbf{x} \\ &= \int (\mathcal{A}\phi)(\mathbf{x}) q_t(\mathbf{x}) d\mathbf{x}. \end{aligned}$$

By the definition of the adjoint operator,

$$\int (\mathcal{A}\phi)(\mathbf{x}) q_t(\mathbf{x}) d\mathbf{x} = \int \phi(\mathbf{x}) (\mathcal{A}^* q_t)(\mathbf{x}) d\mathbf{x}.$$

On the other hand, differentiating under the integral sign gives

$$\frac{d}{dt} \int \phi(\mathbf{x}) q_t(\mathbf{x}) d\mathbf{x} = \int \phi(\mathbf{x}) \frac{\partial q_t}{\partial t}(\mathbf{x}) d\mathbf{x}.$$

Therefore,

$$\int \phi(\mathbf{x}) \left(\frac{\partial q_t}{\partial t}(\mathbf{x}) - (\mathcal{A}^* q_t)(\mathbf{x}) \right) d\mathbf{x} = 0$$

for every sufficiently regular test function ϕ . Hence

$$\frac{\partial}{\partial t} q_t = \mathcal{A}^* q_t,$$

which is the forward, or Fokker–Planck–Kolmogorov, equation. □

Before stating the general case, consider a one-dimensional process with generator $\mathcal{A}\phi(x) = f(x)\nabla\phi(x) + \frac{1}{2}g(x)^2\nabla^2\phi(x)$ (Proposition 2). Then

$$\begin{aligned}\int (\mathcal{A}\phi)(x)p_t(x) dx &= \int \left(f(x)\nabla\phi(x) + \frac{1}{2}g(x)^2\nabla^2\phi(x) \right) p_t(x) dx \\ &= \int f(x)p_t(x)\nabla\phi(x) dx + \frac{1}{2} \int g(x)^2 p_t(x)\nabla^2\phi(x) dx.\end{aligned}$$

Integrating by parts once in the first term and twice in the second, and assuming boundary terms vanish,

$$\begin{aligned}\int (\mathcal{A}\phi)(x)p_t(x) dx &= - \int \phi(x)\nabla(f(x)p_t(x)) dx + \frac{1}{2} \int \phi(x)\nabla^2(g(x)^2 p_t(x)) dx \\ &= \int \phi(x) \left[-\nabla(f(x)p_t(x)) + \frac{1}{2}\nabla^2(g(x)^2 p_t(x)) \right] dx.\end{aligned}$$

Hence,

$$\mathcal{A}^*p_t(x) = -\nabla(f(x)p_t(x)) + \frac{1}{2}\nabla^2(g(x)^2 p_t(x)),$$

and the forward equation from the previous proposition reads

$$\frac{d}{dt}p_t(x) = -\nabla(f(x)p_t(x)) + \frac{1}{2}\nabla^2(g(x)^2 p_t(x)).$$

This is the classical Fokker–Planck equation. The proposition below generalizes it to $\mathbb{R}^d$.

Proposition 5 (Fokker–Planck–Kolmogorov equation). *Suppose that $Q_t(d\mathbf{x}) = p_t(\mathbf{x})d\mathbf{x}$, and define*

$$D(\mathbf{x}) := L(\mathbf{x})L(\mathbf{x})^\top.$$

Assuming that the boundary terms vanish,

$$\frac{d}{dt}p_t(\mathbf{x}) = \mathcal{A}^*p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot (f(\mathbf{x})p_t(\mathbf{x})) + \frac{1}{2}\text{tr} \left\{ \nabla\nabla^\top (D(\mathbf{x})p_t(\mathbf{x})) \right\}.$$

Proof. The forward equation gives $\frac{d}{dt}p_t = \mathcal{A}^*p_t$. To compute the adjoint, integration by parts gives

$$\begin{aligned}\int (\mathcal{A}\phi)(\mathbf{x}) Q_t(d\mathbf{x}) &= \int \nabla_{\mathbf{x}}\phi(\mathbf{x}) \cdot f(\mathbf{x}) p_t(\mathbf{x}) d\mathbf{x} + \frac{1}{2} \sum_{i,j=1}^d \int D_{ij}(\mathbf{x}) \partial_{x_i x_j} \phi(\mathbf{x}) p_t(\mathbf{x}) d\mathbf{x} \\ &= - \int \phi(\mathbf{x}) \nabla_{\mathbf{x}} \cdot (f(\mathbf{x})p_t(\mathbf{x})) d\mathbf{x} + \frac{1}{2} \sum_{i,j=1}^d \int \phi(\mathbf{x}) \partial_{x_i x_j} (D_{ij}(\mathbf{x})p_t(\mathbf{x})) d\mathbf{x}.\end{aligned}$$

Since $\sum_{i,j} \partial_{x_i x_j} (D_{ij}p_t) = \text{tr} \{ \nabla\nabla^\top (Dp_t) \}$, the expression multiplying ϕ is precisely $\mathcal{A}^*p_t$ as stated in the proposition. $\square$

Remark 1. *If the initial condition is random but the dynamics are deterministic, that is,*

$$d\mathbf{x}_t = f(\mathbf{x}_t) dt,$$

then the law of $\mathbf{x}_t$ is still generally nontrivial, since randomness is transported by the deterministic flow. In this case $L = 0$, and the Fokker–Planck–Kolmogorov equation reduces to

$$\frac{d}{dt}p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot (f(\mathbf{x})p_t(\mathbf{x})).$$

This is the Liouville equation, also known as the continuity equation. Thus, even in the absence of diffusion, the forward equation describes the evolution of uncertainty in the state induced by a random initial condition.

Using the one-dimensional Fokker–Planck equation derived above,

$$\frac{d}{dt}p_t(x) = -\nabla(f(x)p_t(x)) + \frac{1}{2}\nabla^2(g(x)^2p_t(x)),$$

we revisit our three running examples.

Example 11 (Fokker–Planck equation for Brownian motion). *For pure Brownian motion (Example 1), $f \equiv 0$ and $g \equiv 1$. Hence, the adjoint of the generator is*

$$\mathcal{A}^*p(x) = \frac{1}{2}\nabla^2p(x),$$

and the Fokker–Planck equation is therefore

$$\frac{d}{dt}p_t(x) = \mathcal{A}^*p_t(x) = \frac{1}{2}\nabla^2p_t(x).$$

Thus, p_t satisfies the heat equation. We already know p_t explicitly: since $B_t \sim \mathcal{N}(0, t)$ (Example 1), p_t is the Gaussian density with variance t , the classical fundamental solution (heat kernel) of the heat equation started from the point mass $p_0 = \delta_0$.

Example 12 (Fokker–Planck equation for the Ornstein–Uhlenbeck process). *For the Ornstein–Uhlenbeck process (Example 2), $f(x) = -\theta x$ and $g \equiv \sigma$. Hence, the adjoint of the generator is*

$$\mathcal{A}^*p(x) = \theta\nabla(xp(x)) + \frac{\sigma^2}{2}\nabla^2p(x),$$

and the Fokker–Planck equation is

$$\frac{d}{dt}p_t(x) = \mathcal{A}^*p_t(x) = \theta\nabla(xp_t(x)) + \frac{\sigma^2}{2}\nabla^2p_t(x).$$

Example 13 (Fokker–Planck equation for Langevin dynamics). *For the Langevin process (Example 3), $f = \nabla \log \pi$ and $g \equiv \sqrt{2}$. Hence, the adjoint of the generator is*

$$\mathcal{A}^*p(x) = -\nabla \cdot (\nabla \log \pi(x) p(x)) + \nabla^2p(x),$$

and the Fokker–Planck equation is

$$\frac{d}{dt}p_t(x) = \mathcal{A}^*p_t(x) = -\nabla \cdot (\nabla \log \pi(x) p_t(x)) + \nabla^2p_t(x).$$

In particular, suppose that π is specified through a sufficiently regular potential V as

$$\pi(x) = \frac{1}{Z}e^{-V(x)}, \quad Z := \int e^{-V(x)} dx < \infty.$$

Then

$$\nabla \log \pi(x) = -\nabla V(x),$$

and therefore

$$\begin{aligned} \mathcal{A}^*p(x) &= \nabla \cdot (\nabla V(x) p(x)) + \nabla^2p(x) \\ &= \nabla \cdot (p(x) \nabla V(x) + \nabla p(x)). \end{aligned}$$

Consequently, the Fokker–Planck equation becomes

$$\frac{d}{dt}p_t(x) = \nabla \cdot (p_t(x) \nabla V(x) + \nabla p_t(x)).$$

2 Langevin Dynamics

2.1 Generator and stationary distribution

In Example 3 we introduced Langevin dynamics directly through the score $\nabla \log \pi$ of a target density π . We now assume $\pi(\mathbf{x}) \propto e^{-\beta U(\mathbf{x})}$ for a potential U and inverse temperature $\beta > 0$, so that $\nabla \log \pi = -\beta \nabla U$; taking $\beta = 1$ recovers the earlier score-based form.

Consider the overdamped time-homogeneous Langevin dynamics

$$d\mathbf{x}_t = -\nabla_{\mathbf{x}} U(\mathbf{x}_t) dt + \sqrt{2\beta^{-1}} d\mathbf{B}_t,$$

where $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is sufficiently regular and $\beta > 0$.

The general formula for the generator and the adjoint now gives the following specialization.

Proposition 6 (Generator and stationary density). *For every sufficiently regular spatial observable $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, the infinitesimal generator is*

$$\mathcal{A}\phi = \langle -\nabla U, \nabla \phi \rangle + \beta^{-1} \Delta \phi.$$

Its adjoint, acting on a sufficiently regular density q , is

$$\mathcal{A}^*q(\mathbf{x}) = \nabla \cdot (\nabla U(\mathbf{x})q(\mathbf{x}) + \beta^{-1} \nabla q(\mathbf{x})).$$

Moreover, provided that

$$Z_\beta := \int e^{-\beta U(\mathbf{x})} d\mathbf{x} < \infty,$$

the density

$$\pi(\mathbf{x}) = \frac{e^{-\beta U(\mathbf{x})}}{Z_\beta}$$

is stationary.

Proof. The expression for $\mathcal{A}$ follows from the general diffusion generator by taking $f = -\nabla U$ and $L = \sqrt{2\beta^{-1}}I$. To compute its adjoint, let q be a sufficiently regular density. Then

$$\begin{aligned} \int (\mathcal{A}\phi)(\mathbf{x})q(\mathbf{x}) d\mathbf{x} &= - \int \nabla U(\mathbf{x}) \cdot \nabla \phi(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} + \beta^{-1} \int \Delta \phi(\mathbf{x}) q(\mathbf{x}) d\mathbf{x} \\ &= \int \phi(\mathbf{x}) \nabla \cdot (\nabla U(\mathbf{x})q(\mathbf{x})) d\mathbf{x} + \beta^{-1} \int \phi(\mathbf{x}) \Delta q(\mathbf{x}) d\mathbf{x} \\ &= \int \phi(\mathbf{x}) \nabla \cdot (\nabla U(\mathbf{x})q(\mathbf{x}) + \beta^{-1} \nabla q(\mathbf{x})) d\mathbf{x}, \end{aligned}$$

where the boundary terms are assumed to vanish. Hence,

$$\mathcal{A}^*q(\mathbf{x}) = \nabla \cdot (\nabla U(\mathbf{x})q(\mathbf{x}) + \beta^{-1} \nabla q(\mathbf{x})).$$

Now let p_t denote the density of the process. By the forward equation,

$$\frac{d}{dt} p_t = \mathcal{A}^* p_t.$$

If π is a stationary density, then $p_t = \pi$ for all t , and hence

$$\frac{d}{dt} p_t = 0.$$

Therefore, a stationary density must satisfy

$$\mathcal{A}^* \pi = 0,$$

that is,

$$\nabla \cdot (\nabla U(\mathbf{x})\pi(\mathbf{x}) + \beta^{-1}\nabla\pi(\mathbf{x})) = 0.$$

A sufficient way to satisfy this equation is to impose the zero-flux condition

$$\nabla U(\mathbf{x})\pi(\mathbf{x}) + \beta^{-1}\nabla\pi(\mathbf{x}) = 0.$$

Dividing by $\pi(\mathbf{x})$ gives

$$\frac{\nabla\pi(\mathbf{x})}{\pi(\mathbf{x})} = -\beta\nabla U(\mathbf{x}),$$

or equivalently,

$$\nabla \log \pi(\mathbf{x}) = -\beta\nabla U(\mathbf{x}).$$

Therefore,

$$\log \pi(\mathbf{x}) = -\beta U(\mathbf{x}) + c,$$

and hence

$$\pi(\mathbf{x}) \propto e^{-\beta U(\mathbf{x})}.$$

Since $Z_\beta < \infty$, normalization gives

$$\pi(\mathbf{x}) = \frac{e^{-\beta U(\mathbf{x})}}{Z_\beta}.$$

Finally,

$$\nabla\pi(\mathbf{x}) = -\beta\pi(\mathbf{x})\nabla U(\mathbf{x}),$$

so that

$$\nabla U(\mathbf{x})\pi(\mathbf{x}) + \beta^{-1}\nabla\pi(\mathbf{x}) = 0.$$

Thus $\mathcal{A}^* \pi = 0$, and the forward equation gives

$$\frac{d}{dt}\pi = 0,$$

so π is stationary. □

2.2 Dirichlet Form

The representation below expresses the generator in the geometry of its stationary measure and identifies its associated Dirichlet form.

TODO: parei aqui

Lemma 1 (Dirichlet form of the Langevin generator). *Let $\pi(\mathbf{x}) = Z_\beta^{-1}e^{-\beta U(\mathbf{x})}$. For smooth test functions ϕ, φ for which boundary terms vanish,*

$$\mathcal{A}\phi = \frac{1}{\beta\pi}\nabla^\top(\pi\nabla\phi),$$

and

$$\langle \mathcal{A}\phi, \varphi \rangle_{\mathcal{L}_\pi^2} = -\frac{1}{\beta} \int \nabla\phi^\top \nabla\varphi \pi d\mathbf{x}.$$

In particular, $\mathcal{A}$ is symmetric and non-positive on this class of functions.

Proof. Since

$$\pi(\mathbf{x}) = Z_\beta^{-1} e^{-\beta U(\mathbf{x})},$$

we have

$$\log \pi(\mathbf{x}) = -\log Z_\beta - \beta U(\mathbf{x}).$$

Taking the gradient gives

$$\nabla \log \pi(\mathbf{x}) = -\beta \nabla U(\mathbf{x}).$$

On the other hand,

$$\nabla \log \pi(\mathbf{x}) = \frac{\nabla \pi(\mathbf{x})}{\pi(\mathbf{x})},$$

and therefore

$$-\nabla U(\mathbf{x}) = \frac{1}{\beta} \frac{\nabla \pi(\mathbf{x})}{\pi(\mathbf{x})}.$$

Recall that the Langevin generator is

$$\mathcal{A}\phi = -\nabla U \cdot \nabla \phi + \frac{1}{\beta} \Delta \phi.$$

Using the previous identity,

$$\begin{aligned} \mathcal{A}\phi &= \frac{1}{\beta} \frac{\nabla \pi}{\pi} \cdot \nabla \phi + \frac{1}{\beta} \Delta \phi \\ &= \frac{1}{\beta \pi} (\nabla \pi \cdot \nabla \phi + \pi \Delta \phi). \end{aligned}$$

By the product rule for the divergence,

$$\nabla \cdot (\pi \nabla \phi) = \nabla \pi \cdot \nabla \phi + \pi \nabla \cdot (\nabla \phi) = \nabla \pi \cdot \nabla \phi + \pi \Delta \phi.$$

Hence,

$$\mathcal{A}\phi = \frac{1}{\beta \pi} \nabla \cdot (\pi \nabla \phi).$$

We now compute the $\mathcal{L}_\pi^2$ inner product. By definition,

$$\langle \mathcal{A}\phi, \varphi \rangle_{\mathcal{L}_\pi^2} = \int (\mathcal{A}\phi)(\mathbf{x}) \varphi(\mathbf{x}) \pi(\mathbf{x}) d\mathbf{x}.$$

Substituting the divergence form of the generator,

$$\begin{aligned} \langle \mathcal{A}\phi, \varphi \rangle_{\mathcal{L}_\pi^2} &= \int \frac{1}{\beta \pi(\mathbf{x})} \nabla \cdot (\pi(\mathbf{x}) \nabla \phi(\mathbf{x})) \varphi(\mathbf{x}) \pi(\mathbf{x}) d\mathbf{x} \\ &= \frac{1}{\beta} \int \varphi(\mathbf{x}) \nabla \cdot (\pi(\mathbf{x}) \nabla \phi(\mathbf{x})) d\mathbf{x}. \end{aligned}$$

Using integration by parts, and assuming that the boundary term vanishes,

$$\int \varphi \nabla \cdot (\pi \nabla \phi) d\mathbf{x} = - \int \nabla \varphi \cdot (\pi \nabla \phi) d\mathbf{x}.$$

Therefore,

$$\begin{aligned} \langle \mathcal{A}\phi, \varphi \rangle_{\mathcal{L}_\pi^2} &= -\frac{1}{\beta} \int \nabla \varphi(\mathbf{x}) \cdot \nabla \phi(\mathbf{x}) \pi(\mathbf{x}) d\mathbf{x} \\ &= -\frac{1}{\beta} \int \nabla \phi(\mathbf{x}) \cdot \nabla \varphi(\mathbf{x}) \pi(\mathbf{x}) d\mathbf{x}. \end{aligned}$$

This is the desired Dirichlet form.

Since the Euclidean inner product is symmetric, we obtain

$$\begin{aligned}\langle \mathcal{A}\phi, \varphi \rangle_{\mathcal{L}_\pi^2} &= -\frac{1}{\beta} \int \nabla \phi \cdot \nabla \varphi \pi \, d\mathbf{x} \\ &= -\frac{1}{\beta} \int \nabla \varphi \cdot \nabla \phi \pi \, d\mathbf{x} \\ &= \langle \phi, \mathcal{A}\varphi \rangle_{\mathcal{L}_\pi^2}.\end{aligned}$$

Thus, $\mathcal{A}$ is symmetric on this class of functions.

Finally, taking $\varphi = \phi$,

$$\langle \mathcal{A}\phi, \phi \rangle_{\mathcal{L}_\pi^2} = -\frac{1}{\beta} \int \|\nabla \phi(\mathbf{x})\|^2 \pi(\mathbf{x}) \, d\mathbf{x} \leq 0,$$

since $\beta > 0$, $\pi(\mathbf{x}) \geq 0$, and $\|\nabla \phi(\mathbf{x})\|^2 \geq 0$. Hence, $\mathcal{A}$ is non-positive. $\square$

The symmetry of $\mathcal{A}$ in $\mathcal{L}_\pi^2$ has important spectral consequences. Under appropriate assumptions on its domain, $\mathcal{A}$ admits a self-adjoint realization (for instance, through the closure of the associated Dirichlet form), and the spectral theorem therefore applies. In particular, $\mathcal{L}_\pi^2$ admits a spectral decomposition adapted to $\mathcal{A}$; when the spectrum is discrete, this reduces to an orthonormal basis of eigenfunctions of $\mathcal{A}$. Since $\mathcal{A}$ is also non-positive, its spectrum is contained in $(-\infty, 0]$, and the corresponding Markov semigroup $P_t = e^{t\mathcal{A}}$ acts diagonally on this spectral decomposition, with each spectral mode decaying at a rate determined by its eigenvalue. This provides the basic mechanism behind spectral analyses of convergence to equilibrium, where the gap between the eigenvalue 0 and the rest of the spectrum controls the relaxation rate.

3 Feynman–Kac Formula

3.1 Terminal-value problems

Let us start by considering the following terminal-value problem for a function $u : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$:

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) + f(x) \frac{\partial}{\partial x} u(t, x) + \frac{1}{2} g(x)^2 \frac{\partial^2}{\partial x^2} u(t, x) = 0, \\ u(T, x) = \phi(x), \end{cases}$$

where f , g , and ϕ are given functions and $T > 0$ is a fixed terminal time.

For any $t \in [0, T]$ and $x \in \mathbb{R}$, consider the process $(x_s)_{s \in [t, T]}$ defined by

$$dx_s = f(x_s) ds + g(x_s) dB_s, \quad x_t = x.$$

That is, the process starts from x at time t and evolves according to the same coefficients f and g appearing in the PDE. In fact, the differential operator in the PDE is precisely the infinitesimal generator of this process,

$$\mathcal{A}\varphi(x) = f(x) \frac{d}{dx} \varphi(x) + \frac{1}{2} g(x)^2 \frac{d^2}{dx^2} \varphi(x),$$

so that the PDE can be written as

$$\frac{\partial}{\partial t} u(t, x) + \mathcal{A}u(t, x) = 0, \quad u(T, x) = \phi(x).$$

The connection with the stochastic process follows by applying Itô's formula to $u(s, x_s)$. Indeed,

$$du(s, x_s) = \left(\frac{\partial}{\partial s} u(s, x_s) + \mathcal{A}u(s, x_s) \right) ds + g(x_s) \frac{\partial}{\partial x} u(s, x_s) dB_s.$$

Since u satisfies the PDE, the drift term vanishes, leaving

$$du(s, x_s) = g(x_s) \frac{\partial}{\partial x} u(s, x_s) dB_s.$$

Integrating from t to T , we obtain

$$u(T, x_T) - u(t, x_t) = \int_t^T g(x_s) \frac{\partial}{\partial x} u(s, x_s) dB_s.$$

Since $x_t = x$ and $u(T, x) = \phi(x)$, this becomes

$$\phi(x_T) - u(t, x) = \int_t^T g(x_s) \frac{\partial}{\partial x} u(s, x_s) dB_s.$$

Taking conditional expectations given $x_t = x$, and using the fact that the Itô integral has zero expectation under suitable integrability conditions, gives

$$\mathbb{E}[\phi(x_T) \mid x_t = x] - u(t, x) = 0.$$

Therefore,

$$u(t, x) = \mathbb{E}[\phi(x_T) \mid x_t = x].$$

Thus, the value of the solution at (t, x) can be obtained by starting the diffusion at $x_t = x$, evolving it until the terminal time T , and averaging the terminal value $\phi(x_T)$ over realizations of the process.

The Feynman–Kac representation also suggests a direct numerical procedure for approximating the solution of the PDE. Consider

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) + f(x) \frac{\partial}{\partial x} u(t, x) + \frac{1}{2} g(x)^2 \frac{\partial^2}{\partial x^2} u(t, x) = 0, \\ u(T, x) = \phi(x). \end{cases}$$

For a fixed pair (t, x) , the Feynman–Kac formula gives

$$u(t, x) = \mathbb{E} [\phi(x_T) \mid x_t = x],$$

where $(x_s)_{s \in [t, T]}$ satisfies

$$dx_s = f(x_s) ds + g(x_s) dB_s, \quad x_t = x.$$

Therefore, $u(t, x)$ can be approximated by Monte Carlo simulation. Let

$$x_T^{(1)}, \dots, x_T^{(N)}$$

be the terminal values of N independent realizations of the SDE, all initialized at $x_t = x$. Then

$$u(t, x) \approx \widehat{u}_N(t, x) := \frac{1}{N} \sum_{n=1}^N \phi(x_T^{(n)}).$$

Thus, evaluating the solution of the backward PDE at a given point (t, x) amounts to simulating the associated stochastic process forward from t to T and averaging the terminal values $\phi(x_T)$. By the law of large numbers,

$$\widehat{u}_N(t, x) \longrightarrow u(t, x)$$

as $N \rightarrow \infty$, under the usual integrability assumptions.

Example 14 (Heat equation). *Consider the backward heat equation*

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) + \frac{1}{2} \frac{\partial^2}{\partial x^2} u(t, x) = 0, \\ u(T, x) = e^{-x^2}. \end{cases}$$

The associated stochastic process is a Brownian motion,

$$dx_s = dB_s, \quad x_t = x.$$

Therefore, by the Feynman–Kac formula,

$$u(t, x) = \mathbb{E} \left[e^{-x_T^2} \mid x_t = x \right].$$

Since

$$x_T = x + B_T - B_t$$

and

$$B_T - B_t \sim \mathcal{N}(0, T - t),$$

we may write

$$x_T = x + \sqrt{T - t} Z, \quad Z \sim \mathcal{N}(0, 1).$$

Therefore,

$$\begin{aligned} u(t, x) &= \mathbb{E} \left[\exp \left(- \left(x + \sqrt{T-t} Z \right)^2 \right) \right] \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp \left(- \left(x + \sqrt{T-t} z \right)^2 - \frac{z^2}{2} \right) dz. \end{aligned}$$

Completing the square in the exponent and evaluating the resulting Gaussian integral gives

$$u(t, x) = \frac{1}{\sqrt{1+2(T-t)}} \exp \left(- \frac{x^2}{1+2(T-t)} \right).$$

Hence, in this case, the Monte Carlo approximation obtained from the Feynman–Kac representation can be compared directly with the exact solution, as shown in Figure 4.

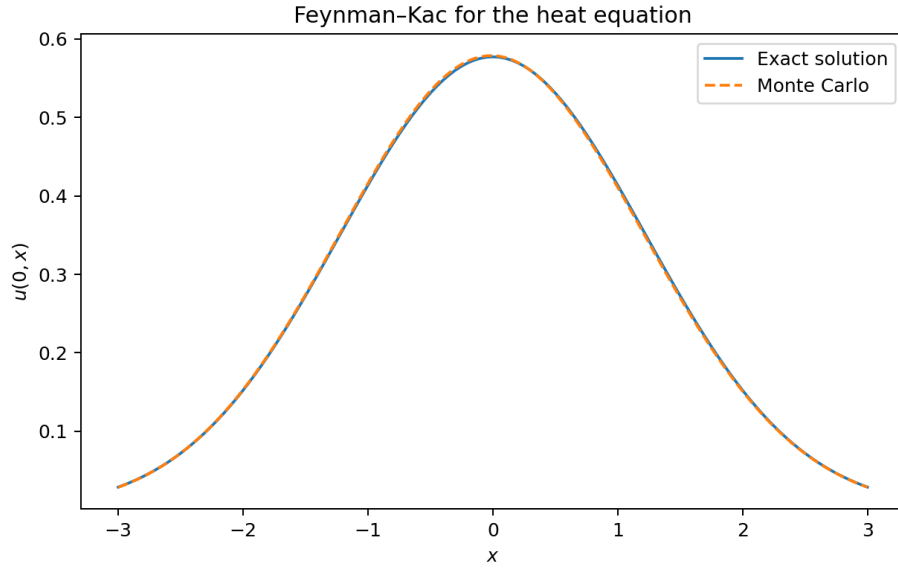


Figure 4: Monte Carlo approximation and exact solution of the backward heat equation.

Example 15 (Ornstein–Uhlenbeck equation). *Consider the backward PDE*

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) - \theta x \frac{\partial}{\partial x} u(t, x) + \frac{\sigma^2}{2} \frac{\partial^2}{\partial x^2} u(t, x) = 0, \\ u(T, x) = x^2. \end{cases}$$

The associated diffusion is the Ornstein–Uhlenbeck process

$$dx_s = -\theta x_s ds + \sigma dB_s, \quad x_t = x.$$

The Feynman–Kac formula gives

$$u(t, x) = \mathbb{E} [x_T^2 \mid x_t = x].$$

Since the Ornstein–Uhlenbeck process has

$$\mathbb{E} [x_T \mid x_t = x] = x e^{-\theta(T-t)}$$

and

$$\text{Var}[x_T \mid x_t = x] = \frac{\sigma^2}{2\theta} \left(1 - e^{-2\theta(T-t)}\right),$$

we obtain

$$u(t, x) = x^2 e^{-2\theta(T-t)} + \frac{\sigma^2}{2\theta} \left(1 - e^{-2\theta(T-t)}\right).$$

The Monte Carlo approximation agrees with the exact solution, as illustrated in Figure 5.

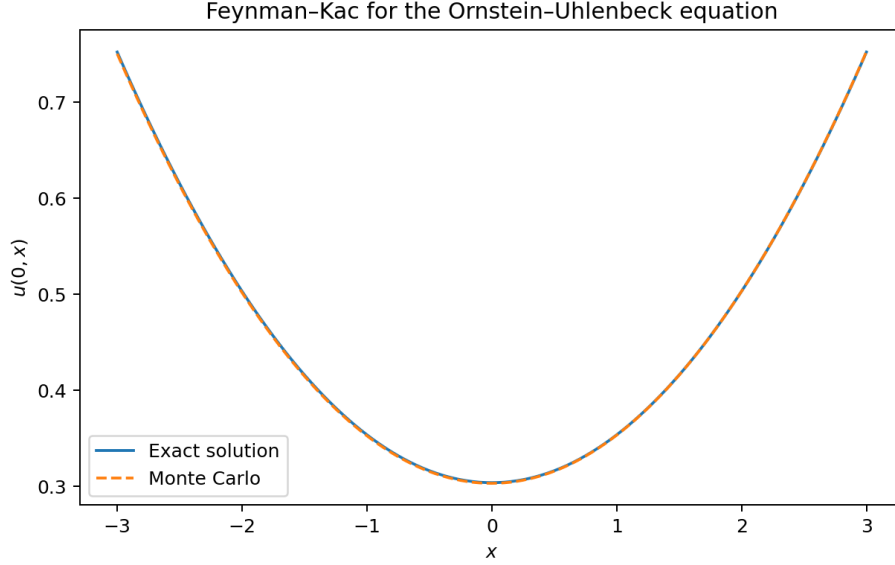


Figure 5: Monte Carlo approximation and exact solution for the Ornstein–Uhlenbeck backward equation.

Example 16 (Feynman–Kac in a two-dimensional double-well potential). *Consider the potential*

$$U(\mathbf{x}) = \frac{1}{4}(x_1^2 - 1)^2 + \frac{1}{2}x_2^2, \quad \mathbf{x} = (x_1, x_2) \in \mathbb{R}^2.$$

Its gradient is

$$\nabla U(\mathbf{x}) = \begin{pmatrix} x_1^3 - x_1 \\ x_2 \end{pmatrix},$$

so the corresponding Langevin diffusion is

$$d\mathbf{x}_s = -\nabla U(\mathbf{x}_s) ds + \sqrt{2\beta^{-1}} d\mathbf{B}_s.$$

Equivalently, componentwise,

$$\begin{cases} dx_{1,s} = (x_{1,s} - x_{1,s}^3) ds + \sqrt{2\beta^{-1}} dB_{1,s}, \\ dx_{2,s} = -x_{2,s} ds + \sqrt{2\beta^{-1}} dB_{2,s}. \end{cases}$$

The potential has two minima, located at $(-1, 0)$ and $(1, 0)$. A realization of the diffusion is shown in Figure 6, superimposed on the level sets of U . The stochastic forcing allows the process to explore the energy landscape and potentially transition between the two wells.

Now consider the backward PDE

$$\begin{cases} \frac{\partial}{\partial t} u(t, \mathbf{x}) + (x_1 - x_1^3) \frac{\partial}{\partial x_1} u(t, \mathbf{x}) - x_2 \frac{\partial}{\partial x_2} u(t, \mathbf{x}) + \beta^{-1} \Delta u(t, \mathbf{x}) = 0, \\ u(T, \mathbf{x}) = \mathbb{I}\{x_1 > 0\}. \end{cases}$$

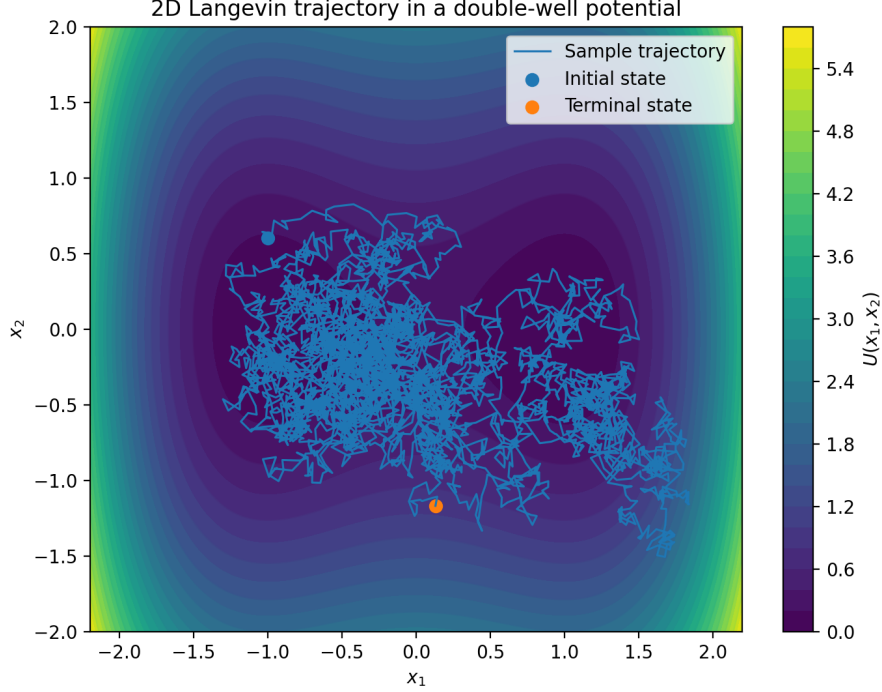


Figure 6: Sample trajectory of the two-dimensional Langevin diffusion superimposed on the level sets of the double-well potential U . The simulation uses $\beta = 1.5$.

The differential operator appearing in this PDE is precisely the infinitesimal generator of the Langevin diffusion above. Therefore, by the Feynman–Kac formula,

$$u(t, \mathbf{x}) = \mathbb{E} [\mathbb{I} \{x_{1,T} > 0\} \mid \mathbf{x}_t = \mathbf{x}].$$

Since the expectation of an indicator function is a probability,

$$u(t, \mathbf{x}) = \mathbb{P}(x_{1,T} > 0 \mid \mathbf{x}_t = \mathbf{x}).$$

Thus, $u(t, \mathbf{x})$ is the probability that a trajectory starting from $\mathbf{x}$ at time t lies in the right-hand well at the terminal time T .

The Feynman–Kac representation also provides a direct numerical approximation of the solution. For a fixed initial condition $\mathbf{x}$, let

$$\mathbf{x}_T^{(1)}, \dots, \mathbf{x}_T^{(N)}$$

be the terminal states of N independent trajectories initialized at $\mathbf{x}_t = \mathbf{x}$. Then

$$u(t, \mathbf{x}) \approx \hat{u}_N(t, \mathbf{x}) := \frac{1}{N} \sum_{n=1}^N \mathbb{I} \{x_{1,T}^{(n)} > 0\}.$$

In other words, $\hat{u}_N(t, \mathbf{x})$ is simply the fraction of simulated trajectories that are in the right-hand well at time T . By the law of large numbers,

$$\hat{u}_N(t, \mathbf{x}) \longrightarrow u(t, \mathbf{x})$$

as $N \rightarrow \infty$, under the usual assumptions.

Repeating this procedure over a grid of initial conditions gives a numerical approximation of the entire solution $u(t, \mathbf{x})$. Figure 7 shows the resulting estimate at $t = 0$. Points initialized sufficiently

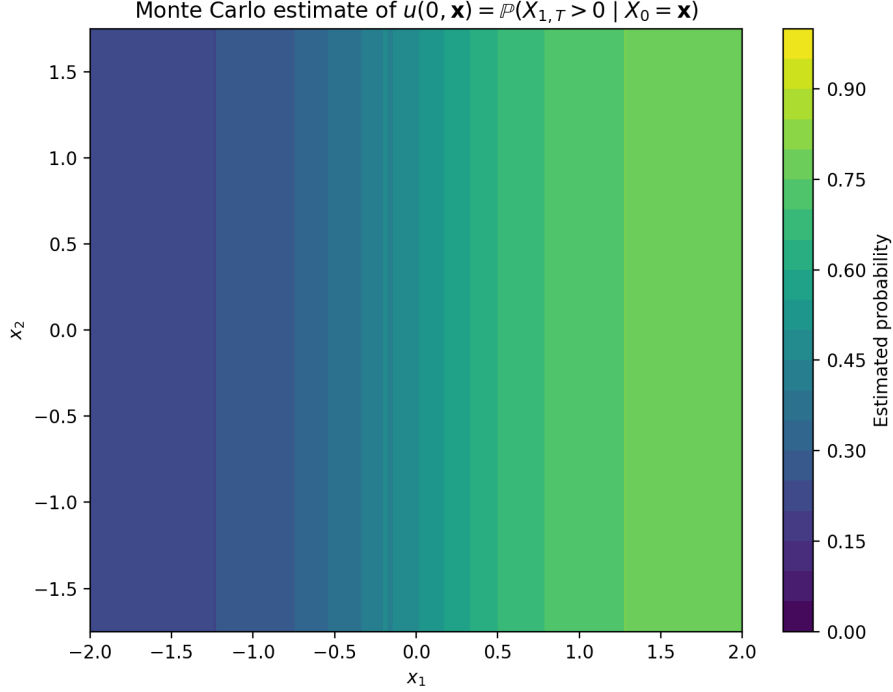


Figure 7: Monte Carlo approximation of $u(0, \mathbf{x}) = \mathbb{P}(x_{1,T} > 0 \mid \mathbf{x}_0 = \mathbf{x})$ for the two-dimensional double-well diffusion, with $\beta = 1.5$ and $T = 1.5$.

far inside the left or right well have probabilities close to 0 or 1, respectively, while initial conditions near the barrier separating the wells have intermediate probabilities.

Notice that, in this particular example, the dynamics of $x_{1,s}$ do not depend on $x_{2,s}$, and the terminal condition also depends only on x_1 . Consequently, $u(t, \mathbf{x})$ is independent of x_2 , which explains the vertical structure observed in Figure 7.

3.2 Boundary value problems and stopping times

The Feynman–Kac formula can also be used to represent solutions of boundary value problems that do not involve a time variable. The main difference from the terminal-value problem considered above is that the diffusion is no longer simulated up to a fixed terminal time T . Instead, it is simulated until it first leaves the domain. This leads naturally to the notion of a *stopping time*, that is, a random time determined by the evolution of the process. We will not develop the general theory of stopping times here, and the argument below should therefore be understood as a formal derivation of the main idea.

Consider the one-dimensional boundary value problem

$$\begin{cases} f(x) \frac{d}{dx} u(x) + \frac{1}{2} g(x)^2 \frac{d^2}{dx^2} u(x) = 0, & x \in D, \\ u(x) = \phi(x), & x \in \partial D, \end{cases}$$

where $D \subset \mathbb{R}$ is a bounded domain and ϕ specifies the value of the solution on its boundary. Recalling that the infinitesimal generator of the diffusion

$$dx_s = f(x_s) ds + g(x_s) dB_s$$

is

$$\mathcal{A}\varphi(x) = f(x) \frac{d}{dx} \varphi(x) + \frac{1}{2} g(x)^2 \frac{d^2}{dx^2} \varphi(x),$$

the PDE in the interior of the domain can be written simply as

$$\mathcal{A}u(x) = 0, \quad x \in D.$$

Fix an initial point $x \in D$ and consider the diffusion

$$dx_s = f(x_s) ds + g(x_s) dB_s, \quad x_0 = x.$$

Let τ denote the first time at which the process exits the domain D . Since the diffusion has continuous trajectories, the corresponding exit point x_τ lies on the boundary ∂D .

Applying Itô's formula to $u(x_s)$ gives

$$\begin{aligned} du(x_s) &= \frac{d}{dx}u(x_s) dx_s + \frac{1}{2} \frac{d^2}{dx^2}u(x_s) (dx_s)^2 \\ &= \frac{d}{dx}u(x_s) (f(x_s) ds + g(x_s) dB_s) + \frac{1}{2} g(x_s)^2 \frac{d^2}{dx^2}u(x_s) ds \\ &= \left[f(x_s) \frac{d}{dx}u(x_s) + \frac{1}{2} g(x_s)^2 \frac{d^2}{dx^2}u(x_s) \right] ds + g(x_s) \frac{d}{dx}u(x_s) dB_s \\ &= (\mathcal{A}u)(x_s) ds + g(x_s) \frac{d}{dx}u(x_s) dB_s. \end{aligned}$$

As long as the process remains inside D , the PDE implies

$$(\mathcal{A}u)(x_s) = 0.$$

Therefore,

$$du(x_s) = g(x_s) \frac{d}{dx}u(x_s) dB_s.$$

We now integrate from the initial time up to the random exit time τ :

$$u(x_\tau) - u(x) = \int_0^\tau g(x_s) \frac{d}{dx}u(x_s) dB_s.$$

At the exit time, $x_\tau \in \partial D$, and hence the boundary condition gives

$$u(x_\tau) = \phi(x_\tau).$$

Thus,

$$\phi(x_\tau) - u(x) = \int_0^\tau g(x_s) \frac{d}{dx}u(x_s) dB_s.$$

Under suitable integrability conditions, the stopped Itô integral has zero expectation. Taking expectations therefore gives

$$\mathbb{E}[\phi(x_\tau)] - u(x) = 0,$$

and consequently

$$u(x) = \mathbb{E}[\phi(x_\tau)].$$

Hence, the value of the solution at an interior point x is the expected boundary value evaluated at the random point where a diffusion initialized at x first exits the domain. In this sense, the boundary condition is transported into the interior through the exit distribution of the stochastic process.

This representation also gives a direct Monte Carlo algorithm for approximating $u(x)$. For a fixed $x \in D$, simulate N independent realizations of the diffusion, all initialized at x , and stop each trajectory when it first exits the domain. If

$$x_\tau^{(1)}, \dots, x_\tau^{(N)}$$

denote the corresponding exit points, then

$$u(x) = \mathbb{E} [\phi(x_\tau)] \approx \hat{u}_N(x) := \frac{1}{N} \sum_{n=1}^N \phi(x_\tau^{(n)}).$$

Thus, solving the boundary value problem at x amounts to simulating the associated stochastic process until it reaches the boundary and averaging the boundary values observed at the random exit locations. By the law of large numbers,

$$\hat{u}_N(x) \longrightarrow u(x)$$

as $N \rightarrow \infty$, under the usual assumptions.

In practice, the SDE is simulated on a discrete time grid. The stopping time τ and the exit point x_τ must therefore also be approximated numerically, for example by detecting the first time step at which the simulated trajectory leaves D .

Example 17 (Steady-state temperature on a circular plate). *Consider the unit disk*

$$D = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}.$$

We study the steady-state temperature $u : D \rightarrow \mathbb{R}$, which satisfies the boundary value problem

$$\begin{cases} \Delta u(\mathbf{x}) = 0, & \mathbf{x} \in D, \\ u(\mathbf{x}) = 1, & \mathbf{x} \in \partial D, x_2 > 0, \\ u(\mathbf{x}) = 0, & \mathbf{x} \in \partial D, x_2 < 0. \end{cases}$$

Thus, the upper semicircle is kept hot and the lower semicircle is kept cold.

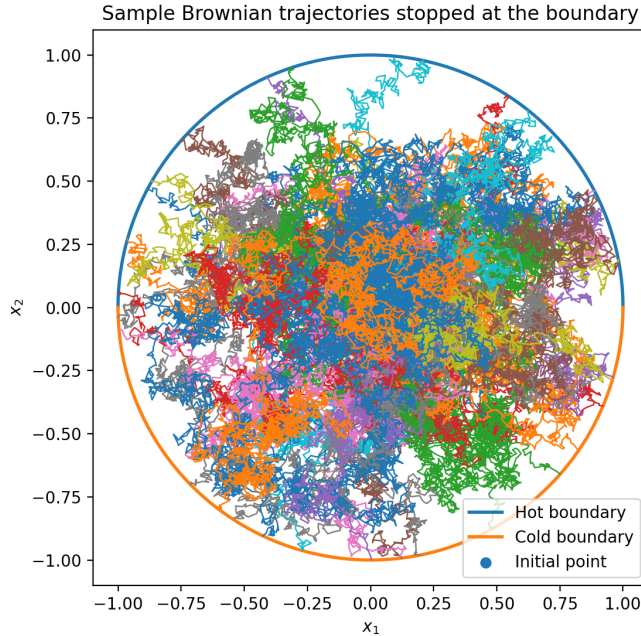


Figure 8: Several Brownian trajectories in the unit disk, stopped at their first exit point on the boundary. The upper semicircle is hot and the lower semicircle is cold.

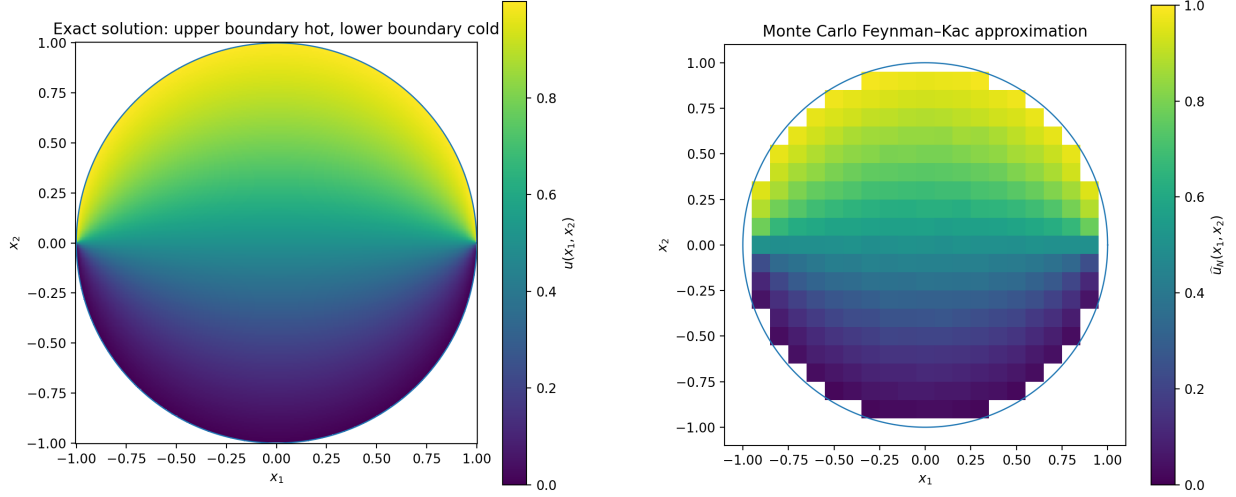
This problem has the explicit solution

$$u(x_1, x_2) = \frac{1}{2} + \frac{1}{\pi} \arctan \left(\frac{2x_2}{1 - x_1^2 - x_2^2} \right).$$

In particular,

$$u(0,0) = \frac{1}{2},$$

and more generally $u(\mathbf{x})$ gives the probability of eventually hitting the hot part of the boundary before the cold part.



(a) Exact solution of the boundary value problem on the disk.

(b) Monte Carlo approximation obtained from the Feynman-Kac representation.

Figure 9: Exact and Monte Carlo solutions for the circular plate problem.

Indeed, let $(\mathbf{x}_s)_{s \geq 0}$ be a two-dimensional Brownian motion started at $\mathbf{x}_0 = \mathbf{x}$, that is,

$$d\mathbf{x}_s = d\mathbf{B}_s.$$

Its infinitesimal generator is

$$\mathcal{A} = \frac{1}{2} \Delta.$$

Since $\Delta u = 0$, we have

$$\mathcal{A}u = 0 \quad \text{in } D.$$

Let τ be the first exit time of the Brownian motion from D . By the Feynman-Kac formula for boundary value problems,

$$u(\mathbf{x}) = \mathbb{E}[\phi(\mathbf{x}_\tau) \mid \mathbf{x}_0 = \mathbf{x}],$$

where

$$\phi(\mathbf{x}) = \mathbb{I}\{x_2 > 0\}, \quad \mathbf{x} \in \partial D.$$

Therefore,

$$u(\mathbf{x}) = \mathbb{E}[\mathbb{I}\{x_{2,\tau} > 0\} \mid \mathbf{x}_0 = \mathbf{x}] = \mathbb{P}(x_{2,\tau} > 0 \mid \mathbf{x}_0 = \mathbf{x}).$$

So the solution $u(\mathbf{x})$ is exactly the probability that a Brownian motion started at $\mathbf{x}$ exits the disk through the hot half of the boundary.

This gives an immediate Monte Carlo approximation. For a fixed initial point $\mathbf{x} \in D$, simulate N independent Brownian trajectories and stop each trajectory at its first exit time from the disk. If

$$\mathbf{x}_\tau^{(1)}, \dots, \mathbf{x}_\tau^{(N)}$$

are the corresponding exit points, then

$$u(\mathbf{x}) \approx \widehat{u}_N(\mathbf{x}) := \frac{1}{N} \sum_{n=1}^N \mathbb{I} \left\{ x_{2,\tau}^{(n)} > 0 \right\}.$$

That is, $\widehat{u}_N(\mathbf{x})$ is simply the fraction of simulated trajectories that exit through the hot half of the boundary.

Figure 8 shows several Brownian trajectories stopped at the boundary. The upper semicircle is hot and the lower semicircle is cold. Figure 9a shows the exact solution, while Figure 9b shows the Monte Carlo approximation obtained from the Feynman–Kac representation.

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